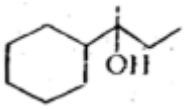
Quizrr Chapter-Wise Test for JEE Advanced - 2025

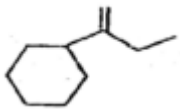
By: C I P H Ξ R

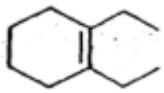
Q.1

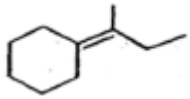
Which of the following is not the product of acidic dehydration of

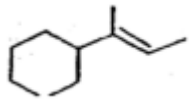


?

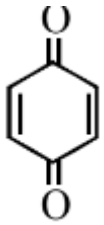
(A) 

(B) 

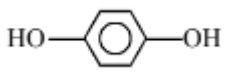
(C) 

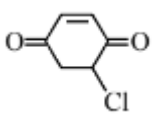
(D) 

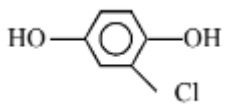
Q.2

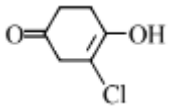


$\xrightarrow{HCl}$ Product is

(A) 

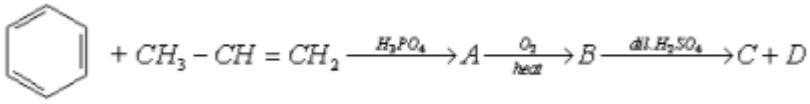
(B) 

(C) 

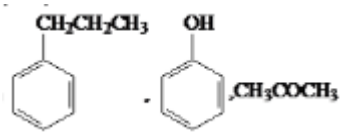
(D) 

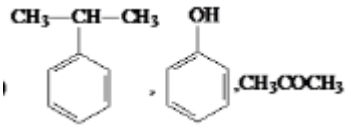
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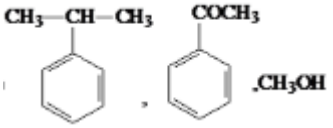
Q.3

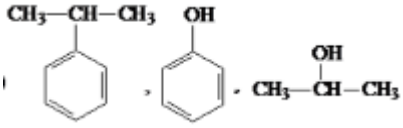


A, C, D are

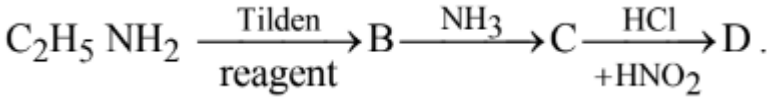
(A) 

(B) 

(C) 

(D) 

Q.4



The product D is an-

(A) Alcohol

(B) Ether

(C) Alkyne

(D) All above

Q.5

What amount of bromine will be required to convert 2 gm of phenol into 2, 4, 6 - tribromophenol -

(A) 4.00

(B) 6.00

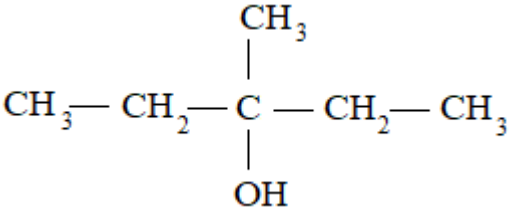
(C) 10.22

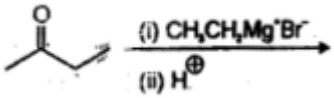
(D) 20.44

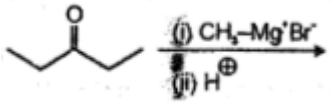
Q.6

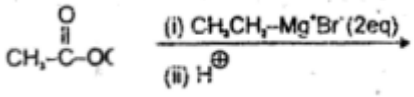
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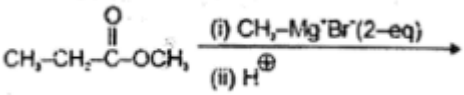
Identify the reaction which can not give 3°-Alcohol



(A) 

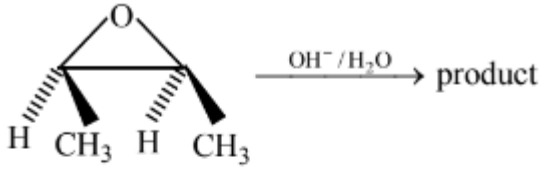
(B) 

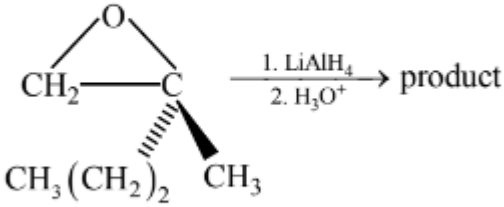
(C) 

(D) 

Q.7

Study following reactions and select incorrect option
(Consider two optical isomers)

I) 

II) 

III) $CH_3CH_2-\underset{\displaystyle OH}{\underset{|}{CH}}-COOH \xrightarrow{\Delta} \text{product}$

IV) $CH_3-CH_2-CH_2-\underset{\displaystyle OH}{\underset{|}{CH}}-CH_2-COOH \xrightarrow{\Delta} \text{product}$

(A) In the I^{st} reaction product is a mixture of two optical isomers.

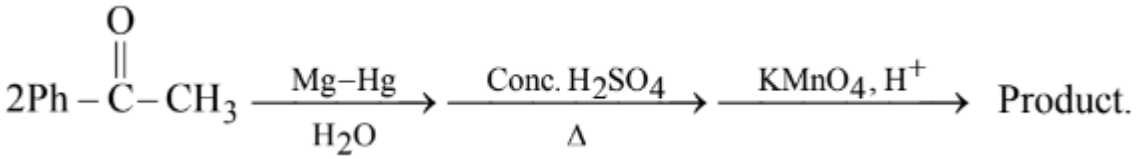
(B) In the I^{nd} reaction product is only one optical isomer.

(C) In the III rd reaction product is a mixture of two optical isomers.

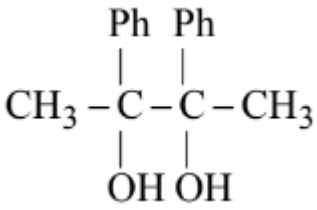
(D) In IVth reaction total no. of products two

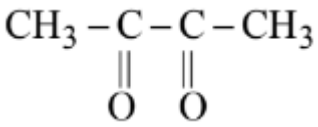
Q.8

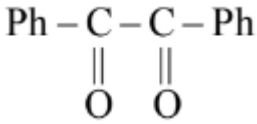
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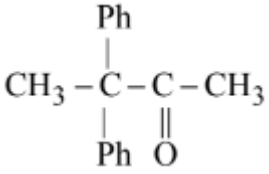


The final product is -

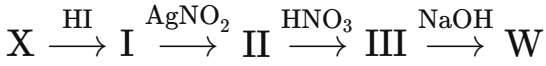
(A) 

(B) 

(C) 

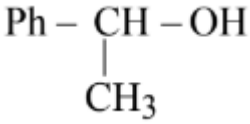
(D) 

Q.9

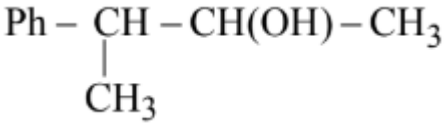


Choose the correct combination of X and W

(A) X - $CH_3 - CH = CH - OH$, W (Colour of final product) - Red

(B) X - 

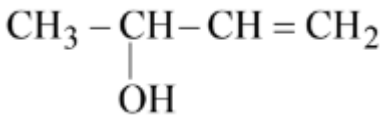
W (Colour of final product) - Blue

(C) X - 

W (Colour of final product) - White

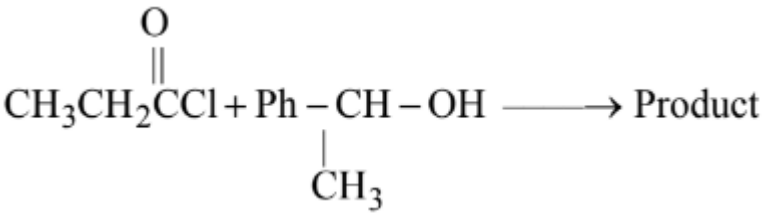
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(D) X -



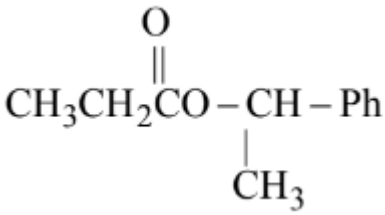
W (Colour of final product) - Red

Q.10

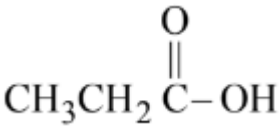


The main product is/are -

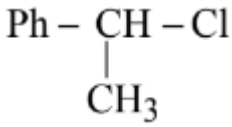
(A)



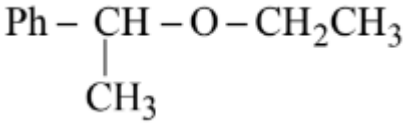
(B)



(C)

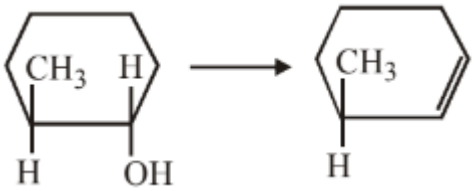


(D)



Q.11

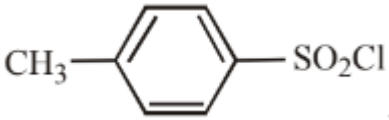
Which of the following represents condition(s) to perform the given conversion?



(A) Na-metal; CS_2; heat

(B) $POCl_3$, pyridine

(C)



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, pyridine; CH, CH_3CH_2OH

(D) CF_3SO_2Cl, pyridine, $Me_3C-O^-K^+, Me_3C-OH$

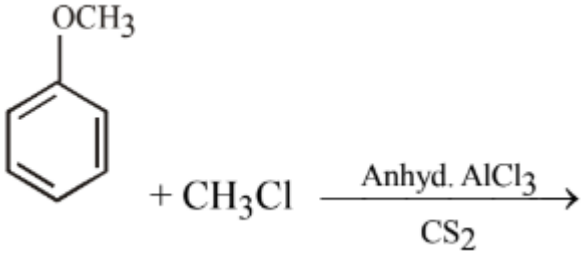
Q.12

Passage:

Anisole undergoes Friedel-Crafts reaction, i.e., the alkyl and acyl groups are introduced at ortho and para positions by reaction with alkyl halide and acyl halide in the presence of anhydrous aluminium chloride (a Lewis acid) as catalyst. Anisole reacts with a mixture of concentrated sulphuric and nitric acids to yield a mixture of ortho and para nitroanisole.

Question:

In reaction



The major product is -

(A) 2-Methoxy toluene

(B) 4-Methoxy toluene

(C) 3-Methoxy toluene

(D) None of these

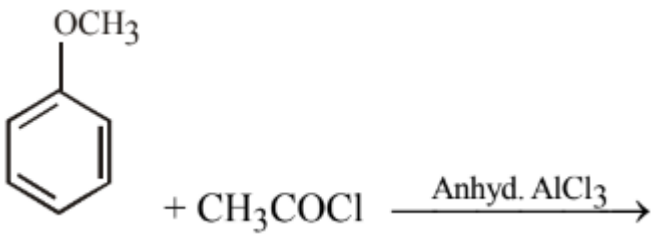
Q.13

Passage:

Anisole undergoes Friedel-Crafts reaction, i.e., the alkyl and acyl groups are introduced at ortho and para positions by reaction with alkyl halide and acyl halide in the presence of anhydrous aluminium chloride (a Lewis acid) as catalyst. Anisole reacts with a mixture of concentrated sulphuric and nitric acids to yield a mixture of ortho and para nitroanisole.

Question:

In reaction



The minor product is -

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(A) 2-Methoxy acetophenone

(B) 3-Methoxy acetophenone

(C) 4-Methoxy acetophenone

(D) None of these

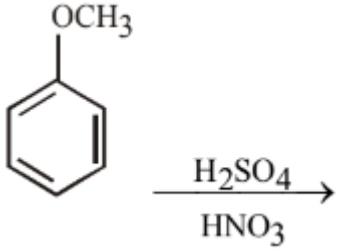
Q.14

Passage:

Anisole undergoes Friedel-Crafts reaction, i.e., the alkyl and acyl groups are introduced at ortho and para positions by reaction with alkyl halide and acyl halide in the presence of anhydrous aluminium chloride (a Lewis acid) as catalyst. Anisole reacts with a mixture of concentrated sulphuric and nitric acids to yield a mixture of ortho and para nitroanisole.

Question:

In reaction



The major prodcut is -

(A) 4-Nitroanisole

(B) 2-Nitroanisole

(C) 3-Nitroanisole

(D) None of these

Q.15

Passage:

Diane Diol had worked for several days to prepare the compounds shown. She carefully labelled them and went to launch. To her horror, when she returned she found that the labels had fallen off the bottles onto the floor. Gladys Glycol, the student at the next bench, told her that the diols could be easily distinguished by two experiments. All Dioane had to do was to determine which ones were optically active and how many products were obtained when each was treated with a periodic acid. Diane did what Gladys suggested and found the following :

(1) Compounds A, E and F were optically active B, C and D were optically inactive.

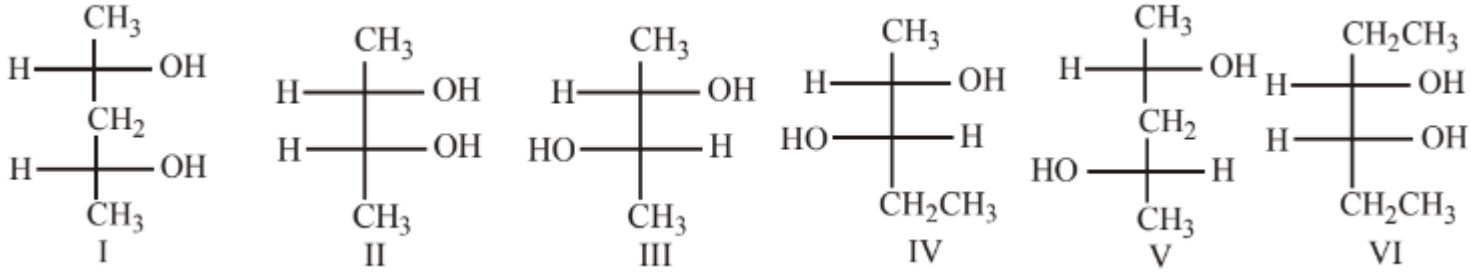
(2) One product was obtained from the reaction A, B and D with the periodic acid.

(3) Two products were obtained from the reaction of F with the periodic acid.

(4) C and E didn't react with the periodic acid.

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The structures of the compound were :



Question:

On the basis of observation, which structure could be suggested as compound A ?

(A) II

(B) III

(C) IV

(D) VI

Q.16

Passage:

Diane Diol had worked for several days to prepare the compounds shown. She carefully labelled them and went to launch. To her horror, when she returned she found that the labels had fallen off the bottles onto the floor. Gladys Glycol, the student at the next bench, told her that the diols could be easily distinguished by two experiments. All Dioane had to do was to determine which ones were optically active and how many products were obtained when each was treated with a periodic acid. Diane did what Gladys suggested and found the following :

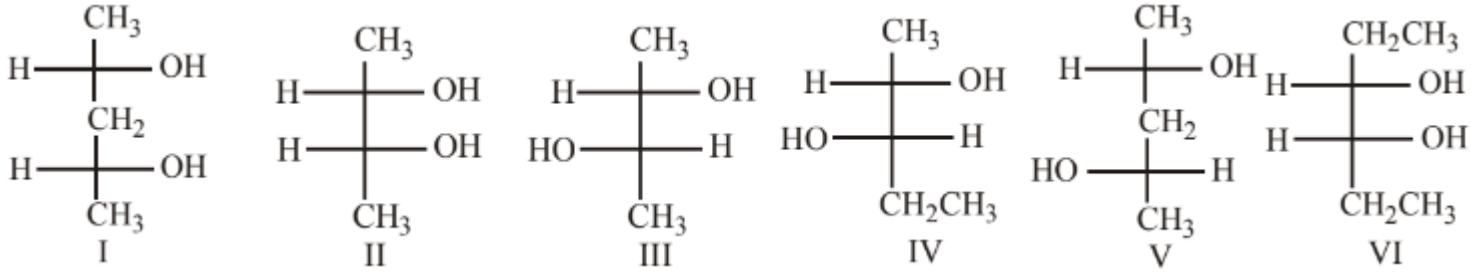
(1) Compounds A, E and F were optically active B, C and D were optically inactive.

(2) One product was obtained from the reaction A, B and D with the periodic acid.

(3) Two products were obtained from the reaction of F with the periodic acid.

(4) C and E didn't react with the periodic acid.

The structures of the compound were :



Question:

On the basis of observation, which structure could be suggested as compound F ?

(A) II

(B) III

(C) IV

(D) VI

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Q.17

Passage:

Diane Diol had worked for several days to prepare the compounds shown. She carefully labelled them and went to launch. To her horror, when she returned she found that the labels had fallen off the bottles onto the floor. Gladys Glycol, the student at the next bench, told her that the diols could be easily distinguished by two experiments. All Dioane had to do was to determine which ones were optically active and how many products were obtained when each was treated with a periodic acid. Diane did what Gladys suggested and found the following :

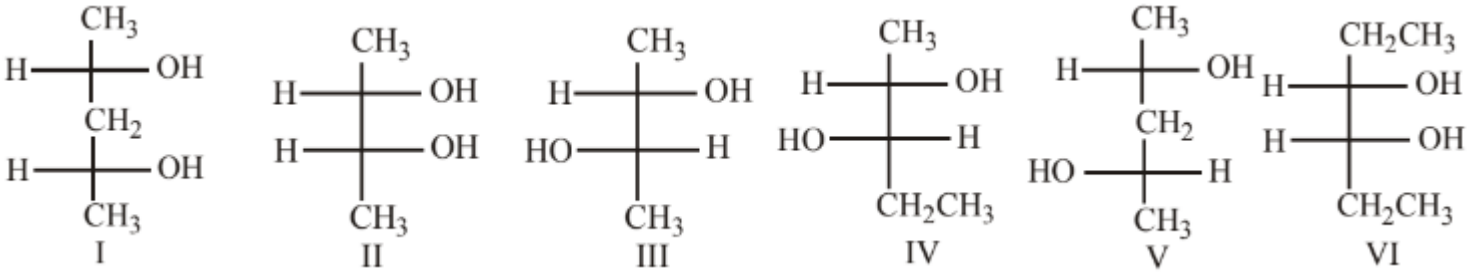
(1) Compounds A, E and F were optically active B, C and D were optically inactive.

(2) One product was obtained from the reaction A, B and D with the periodic acid.

(3) Two products were obtained from the reaction of F with the periodic acid.

(4) C and E didn't react with the periodic acid.

The structures of the compound were :



Question:

On the basis of observation, which structure could be suggested as compound E ?

(A) I

(B) III

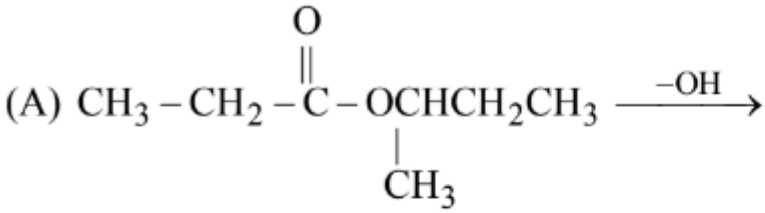
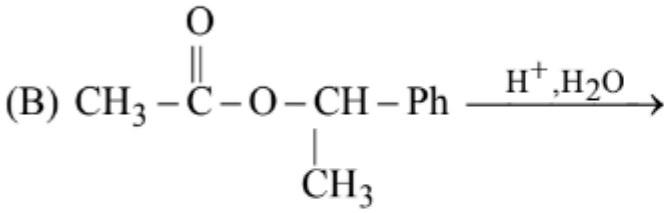
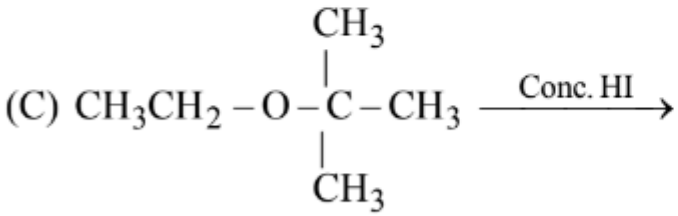
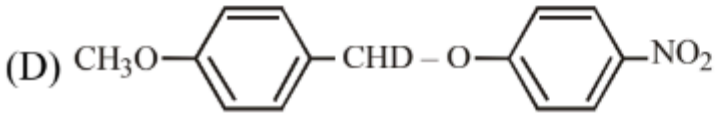
(C) IV

(D) V

Q.18

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Match the following:

Column I	Column II
(A) 	(p) Involves carbocation formation
(B) 	(q) Acyl cleavage
(C) 	(r) Racemic mixture
(D) 	(s) Product gives iodoform test

(A) (A) - r,s, (B) - q,p, (C) - r, (D) - s,p

(B) (A) - p,s, (B) - q,s, (C) - q,p, (D) - r

(C) (A) - q,s, (B) - p,r, (C) - p,s, (D) - p,r

(D) (A) - p, (B) - q,s, (C) - p,s, (D) - q,p

Q.19

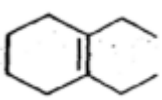
An organic compound (A) contains only C, H and O atoms. On reaction with excess of CH_3MgI, (A) gives a gas (X), whose volume under STP conditions was found to be 67.2 litre. What is/are the number of hydroxyl group (s) in the compound (A)?

Q.20

A compound with molecular formula $C_6H_{14}O_5$ is converted by the action of acetyl chloride to a compound with molecular mass 334 . The number of OH groups in the compound is .....

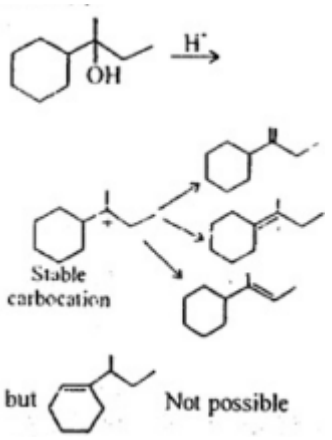
Answers & Solutions

Q.1 Answer:

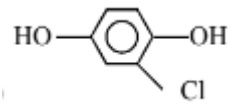


Solution:

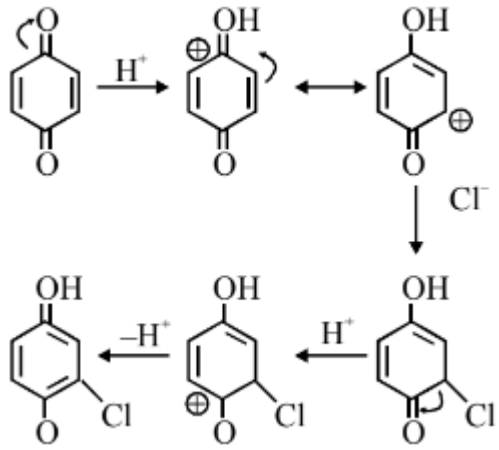
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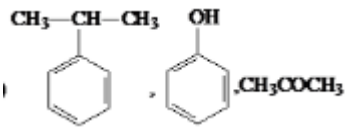
Q.2 Answer:



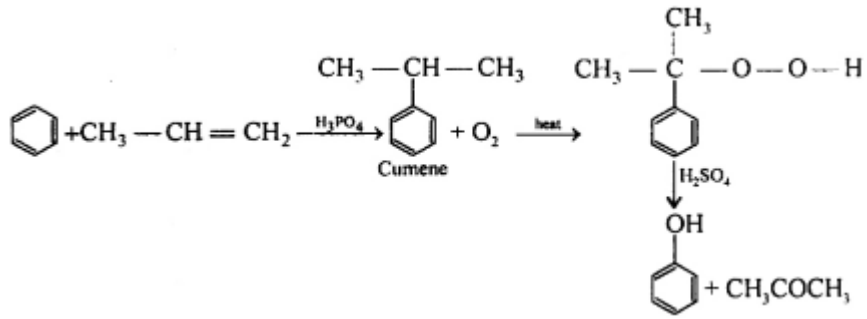
Solution:



Q.3 Answer:



Solution:

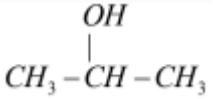


(A) is not possible because $CH_3 - CH_2 - \overset{\oplus}{C}H_2$ **is less stable than** $CH_3 - \overset{\oplus}{C}H - CH_3$

(C) is not possible because acetophenone and CH_3OH **cannot be formed.**

(D) is not possible because

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is formed here.

Q.4 Answer:

Alcohol

Solution:

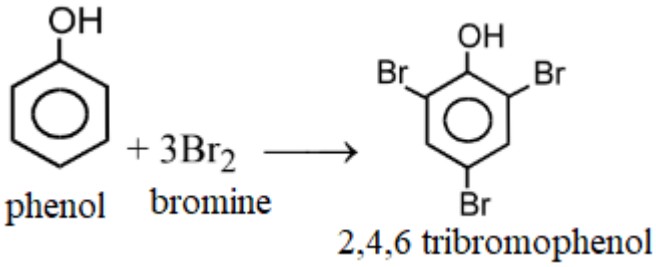
In the above reaction the product D is C_2H_5OH (Ethanol) The reaction takes place as follows :

$$C_2H_5NH_2 \xrightarrow[-N_2,-H_2O]{\text{Tilden reagent (NOCl)}} C_2H_5Cl \xrightarrow{+NH_3} C_2H_5NH_2 \xrightarrow[-N_2-HCl]{+HNO_2/HCl} C_2H_5OH$$

Q.5 Answer:

10.22

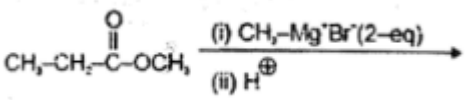
Solution:



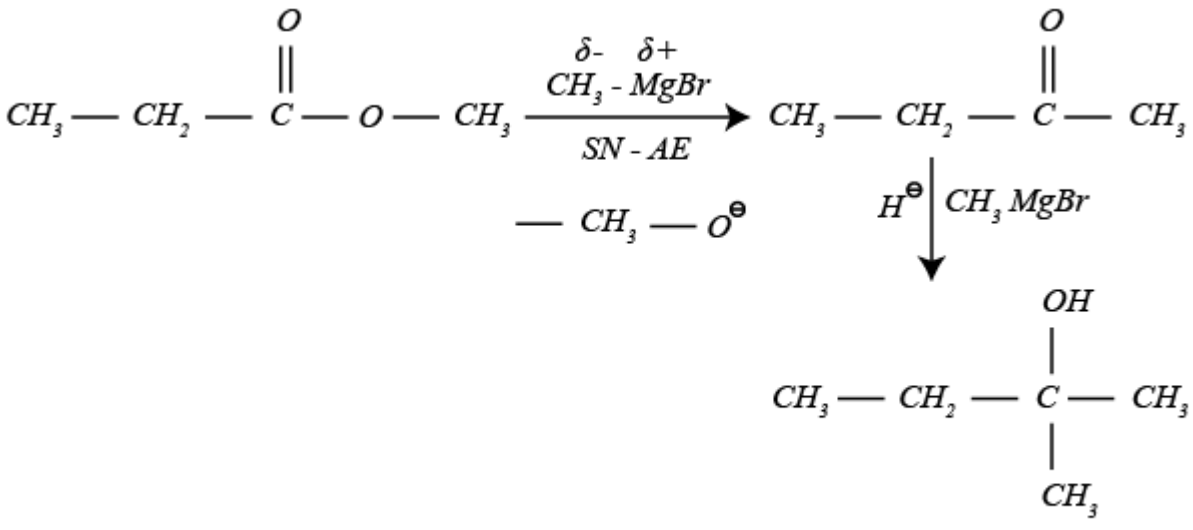
94 g **of phenol requires** 480 gBr_2

2 g **of phenol requires** $= \frac{480}{94} \times 2 = 10.22$ g **bromine**

Q.6 Answer:



Solution:



By nucleophilic addition reaction

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Q.7 Answer:

In the III rd reaction product is a mixture of two optical isomers.

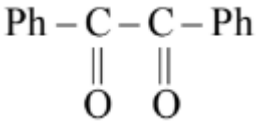
Solution:

I - Resolvable racemic mixture

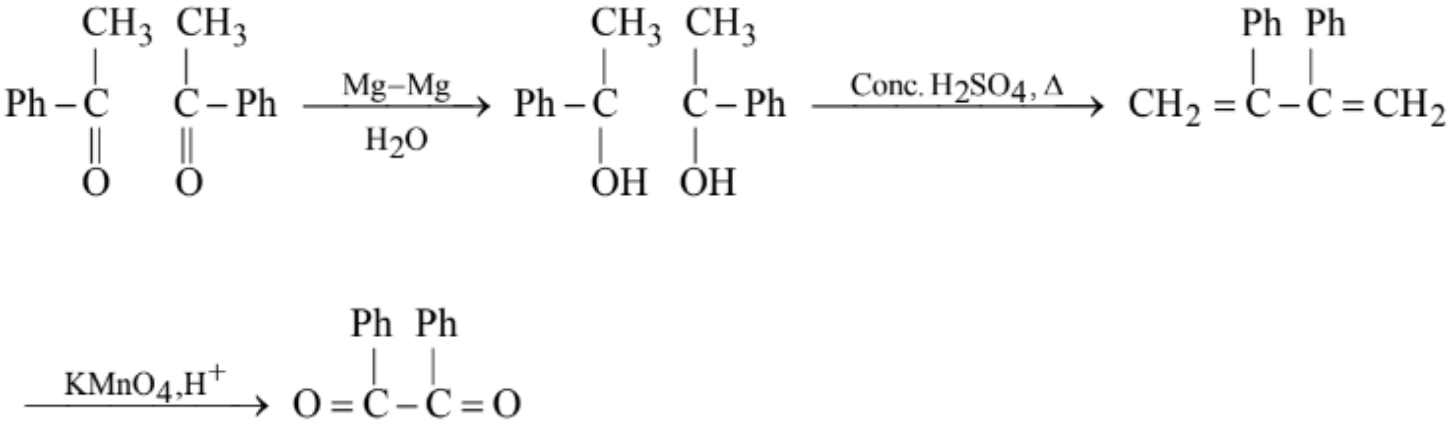
II - Optically active product due to SN^2 attack only one product III - Three optical isomers are possible out of which one is meso compound due to centre of symmetry and other two comprising a racemic mixture.

IV - Two product cis & trans

Q.8 Answer:

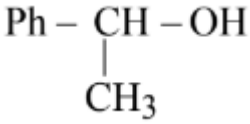


Solution:



Q.9 Answer:

X -

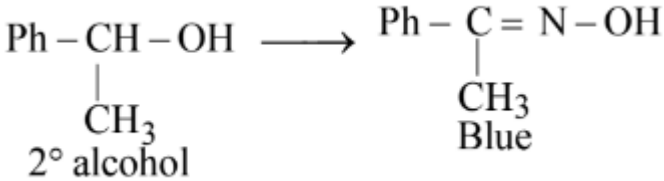


W (Colour of final product) - Blue

Solution:

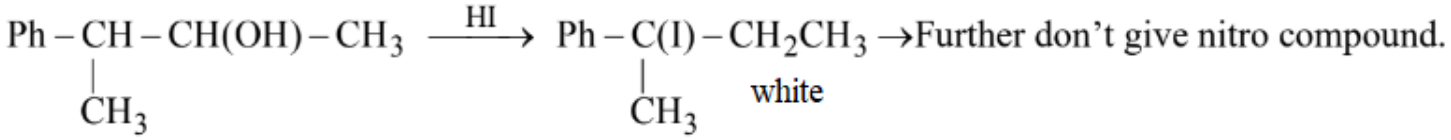
(A) $CH_3CH = CHOH \xrightarrow{HI}$ **don't give idodie.**

(B)

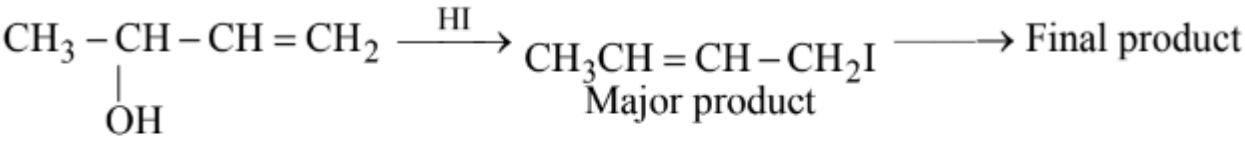


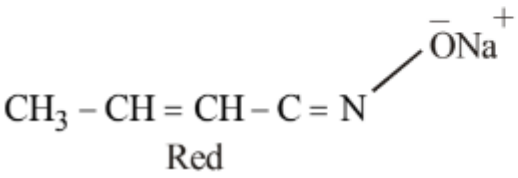
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(C)

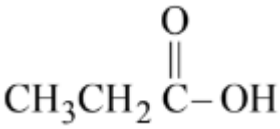


(D)

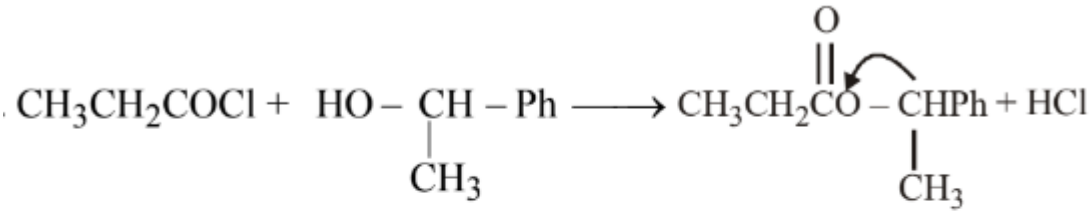


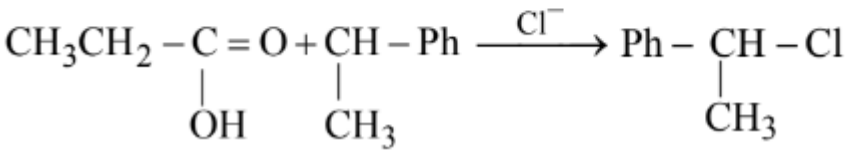


Q.10 Answer:



Solution:





Q.11 Answer:

Na-metal; CS_2; heat

Solution:

(A) Provides pyrolysis of xanthane (less substituted alkene is pereferred)

(B) $POCl_3$**, pyridine follows** E_2 **path**

(C) and (D) also follow the E_2 **path.**

E_2 **proceeds through anti-elimination.**

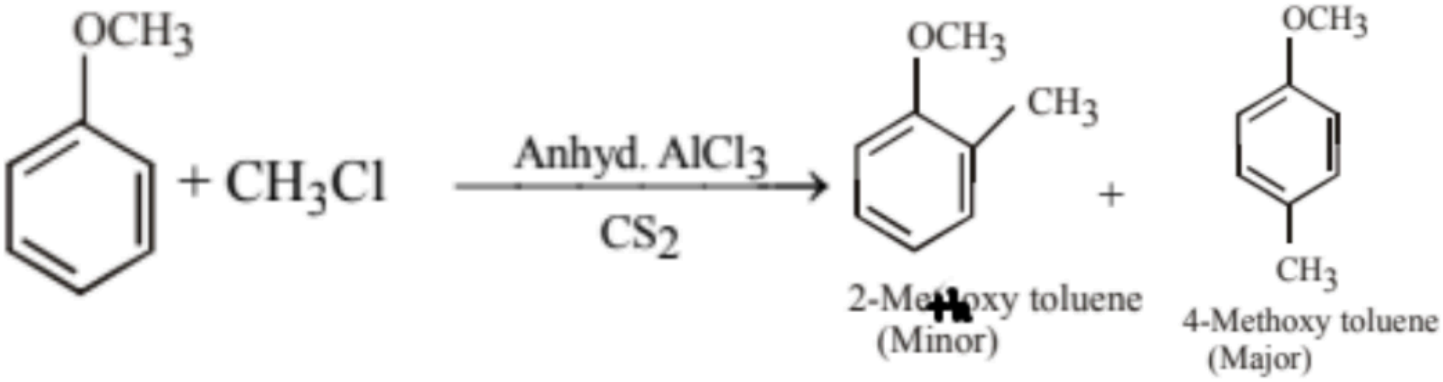
Q.12 Answer:

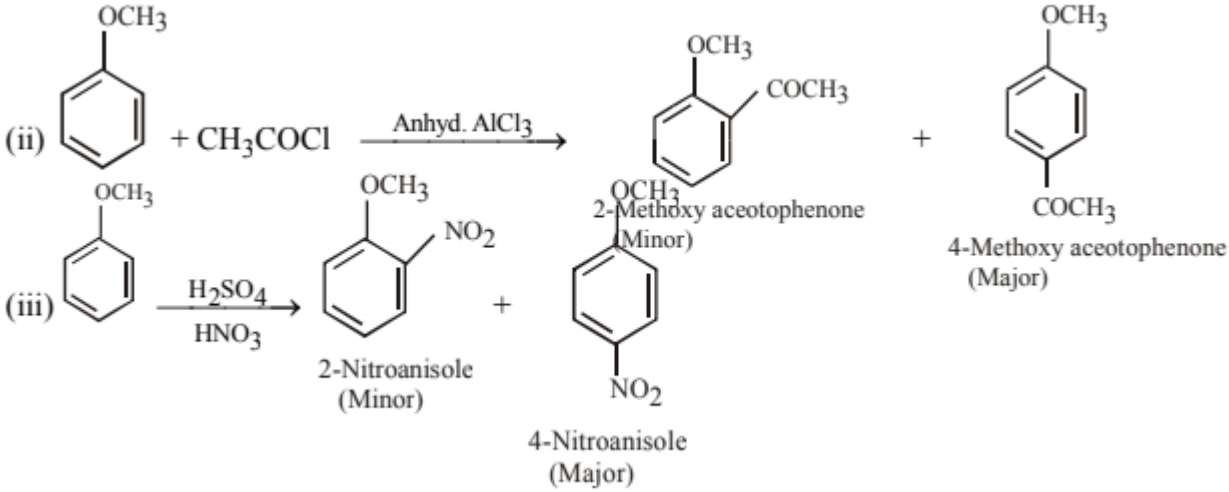
4-Methoxy toluene

Solution:

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(i)



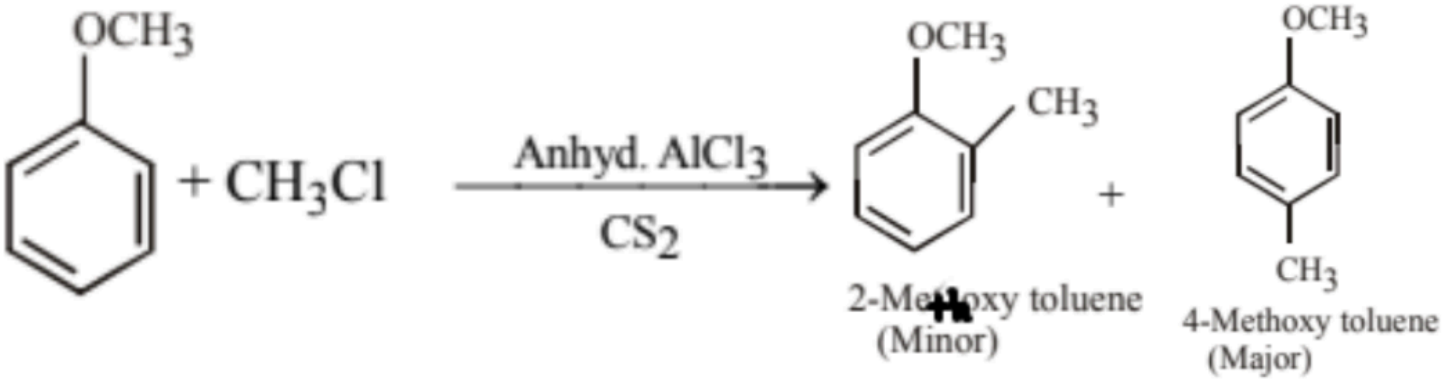


Q.13 Answer:

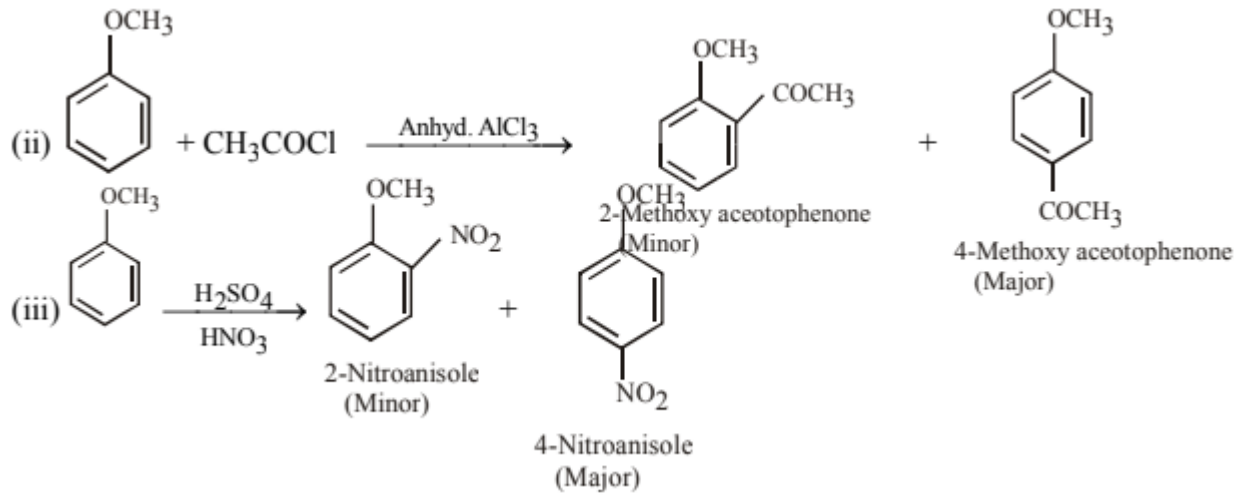
2-Methoxy acetophenone

Solution:

(i)



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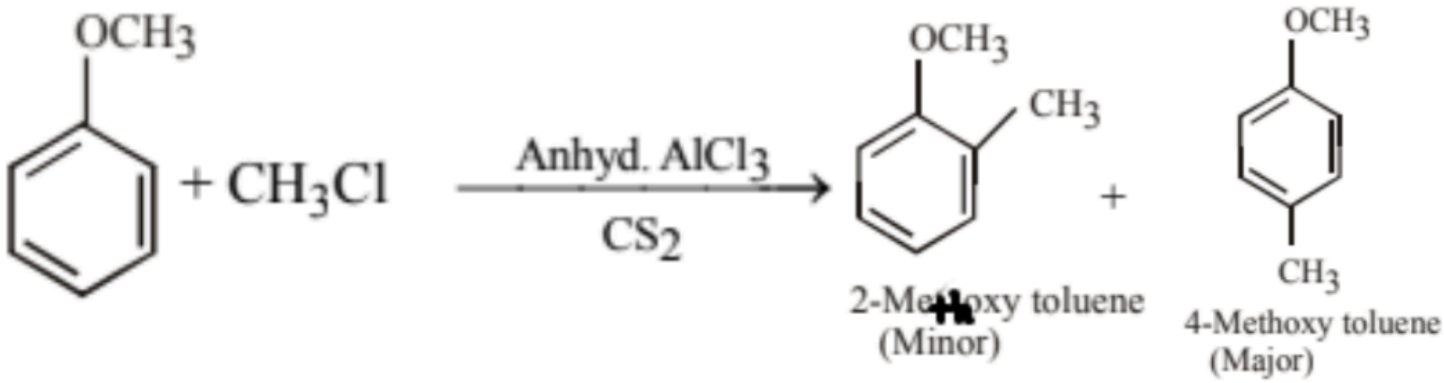


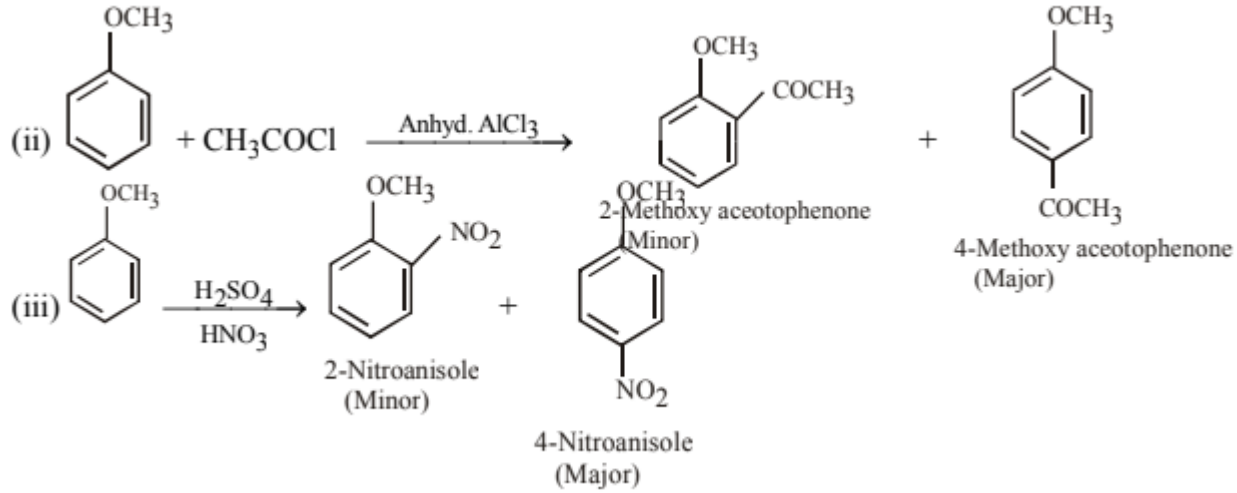
Q.14 Answer:

4-Nitroanisole

Solution:

(i)





Q.15 Answer:

III

Solution:

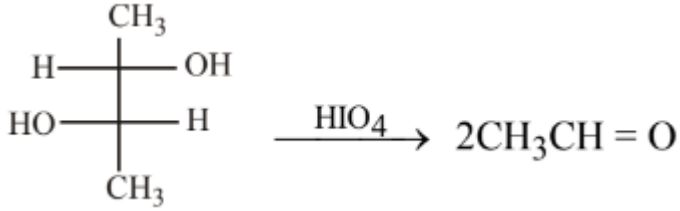
(i) Regarding A, the passage information :

∗ Optically active

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∗ One product is obtained with HIO_4.

Hence, A is the structure III

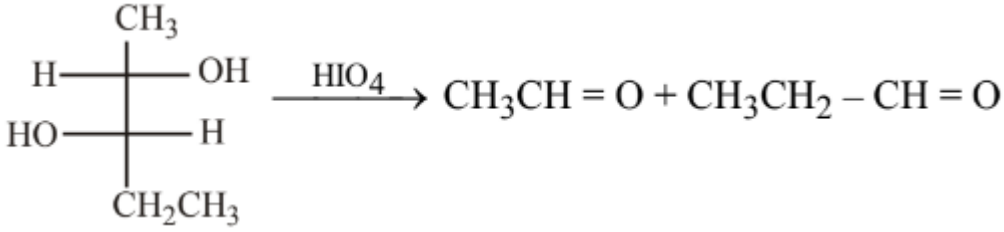


(ii) For compound F, the passage information :

∗ Optically active

∗ Two products are obtained with HIO_4.

Hence, F is the structure IV



(iii) For compound E, the passage information:

∗ Optically active

∗ no reaction with HIO_4.

Hence, E is the structure V

Q.16 Answer:

IV

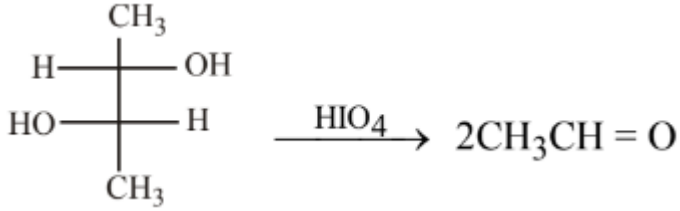
Solution:

(i) Regarding A, the passage information :

∗ Optically active

∗ One product is obtained with HIO_4.

Hence, A is the structure III



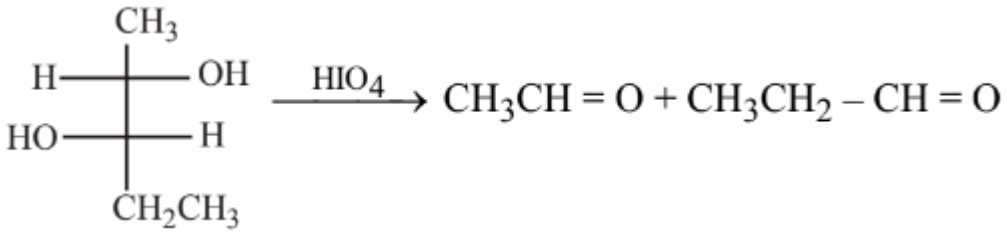
(ii) For compound F, the passage information :

∗ Optically active

∗ Two products are obtained with HIO_4.

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Hence, F is the structure IV



(iii) For compound E, the passage information:

∗ Optically active

∗ no reaction with HIO_4.

Hence, E is the structure V

Q.17 Answer:

V

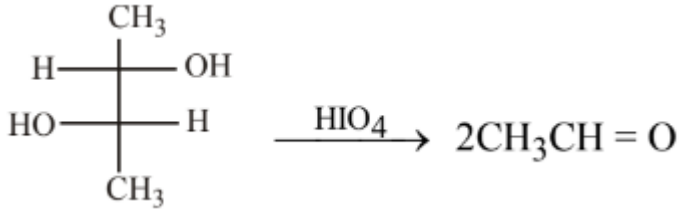
Solution:

(i) Regarding A, the passage information :

∗ Optically active

∗ One product is obtained with HIO_4.

Hence, A is the structure III

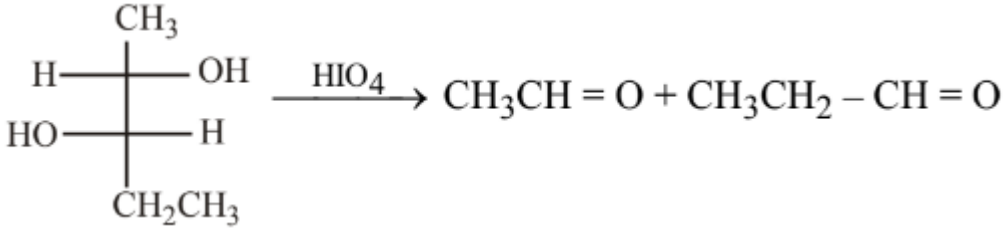


(ii) For compound F, the passage information :

∗ Optically active

∗ Two products are obtained with HIO_4.

Hence, F is the structure IV



(iii) For compound E, the passage information:

∗ Optically active

∗ no reaction with HIO_4.

Hence, E is the structure V

Q.18 Answer:

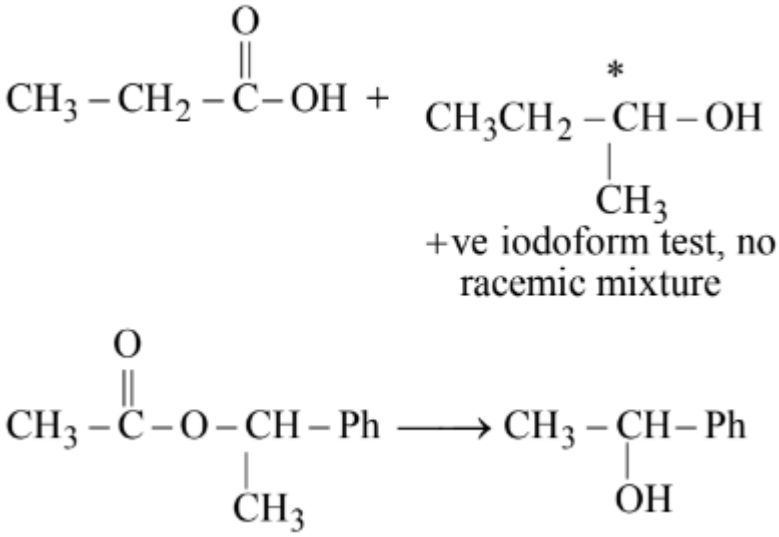
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(A) - q,s, (B) - p,r, (C) - p,s, (D) - p,r

Solution:

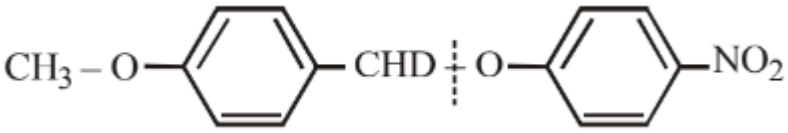
$(A) - qs, (B) - pr, (C) - ps, (D) - rp$

A – the same product



Carbocation forms; Racemisation occurs; gives iodoform test.

$CH_3 - CH_2 - O - CMe_3$; **Carbocation forms but no recemisation,** $CH_3CH_2 - OH$ **gives iodoform test.**



Carbocation forms, recemisation, occurs.

Q.19 Answer:

—

Solution:

$\frac{67.2}{22.4}$ **= moles of gas** (X) **evolved** $= 3$ **mole of gas** (X)

The number of active H-atom present in a molecule is equal to the number of moles of gas obtained i.e. number of $-OH$ **groups in compound** $(A) = 3$.

Q.20 Answer:

—

Solution:

$R - OH + CH_3COCl \longrightarrow R - OCOCH_3 + HCl$

When an alcohol containing one OH **group reacts with 1 mole of** CH_3COCl**, the molecular mass of the product will be 42 units more than that of** ROH**. If the original compound contains '** n **'** OH **groups, then the molar mass of product will be** $42n$ **units more than that of original compound. In the present case,**

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$166 + 42n = 334 \therefore n = 4$

Thus, the compound contains $4 - OH$ groups.

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